CHAPTER 06

Electromagnetic Induction

In this Chapter...

- Faraday's Laws and Motional Electromotive Force
- Self and Mutual Induction

In 1820, Hans Christian Oersted discovered that moving charges or electric current produces magnetic field. After this discovery, the question arises, if electric current produces magnetic field, then is it possible to produce electric current due to magnetic field.

To demonstrate this reverse effect, Michael Faraday in England and Joseph Henry in USA around 1830 carried out experiments and observed that an electric current can be induced in a closed coil, when it is subjected to change in magnetic field.

The phenomenon of generation of induced current in a coil due to change in magnetic field through the coil is called **Electromagnetic Induction** (EMI).

Power generator, dynamos, transformers etc. are based upon the electromagnetic induction.

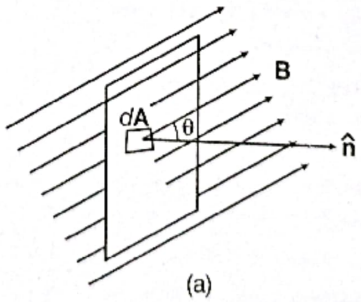
Faraday's Laws and Motional Electromotive Force

The phenomenon of generation of current or emf by changing the magnetic field is known as **Electromagnetic Induction** (EMI). The emf developed in the conductor by the process of EMI is known as **induced emf** and if the conductor is in the form of a closed loop, then the current flowing through the conductor is known as **induced current**.

It is the reverse process of magnetic field production by electric current.

Magnetic Flux

The total number of magnetic field lines crossing through any surface normally, when it is placed in a magnetic field is known as the magnetic flux of that surface.



Suppose a loop enclosing an area **A** is placed in a uniform magnetic field **B**. Then, the magnetic flux through the loop is given by

$$\phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

- When the magnetic field is perpendicular to that plane of the loop, then magnetic flux will be

$$\phi_B = BA = \text{maximum value} \quad \ldots(i)$$

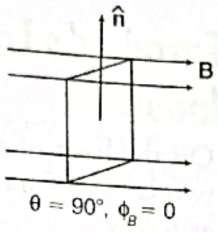
This means that $B = \frac{\phi_B}{A}$, i.e. magnetic field strength B is the magnetic flux per unit area and is called **magnetic flux density** or **magnetic induction.**

- When the magnetic field B is not perpendicular to area A rather it is inclined at an angle θ with respect to the normal to the surface, then the magnetic flux becomes

$$\phi_B = \mathbf{B}\cdot\mathbf{A} = |\mathbf{B}||\mathbf{A}|\cos\theta = BA\cos\theta \quad ...(ii)$$

where, θ is the smaller angle between **B** and **A**.

- If a plane is parallel to the magnetic field, then no field line will pass through it and the magnetic flux linked with that plane will be zero.



From Eq.(ii), it is clear that the flux can be varied by changing anyone or more of the terms **B**, **A** and θ.

Hence, the flux can be altered by changing the shape of the coil (by stretching or shrinking) in a magnetic field or rotating a coil in a magnetic field, such that the angle θ between **B** and **A** changes.

Units and Dimensional formula

- The SI unit of magnetic flux (ϕ_B) is tesla-metre square, which is also called **weber** (abbreviated Wb).

$$1\ \text{Wb} = 1\ \text{T-m}^2$$

1 weber is the amount of magnetic flux over an area of 1m^2 held normal to a uniform magnetic field of 1 tesla (T).

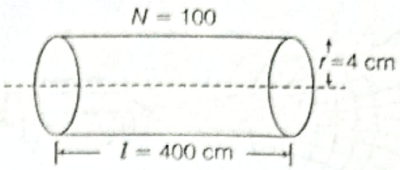
- The CGS unit of magnetic flux (ϕ_B) is maxwell (Mx).

where, $1\ \text{weber} = 10^8\ \text{maxwell}$

Magnetic flux is a scalar quantity and its dimensional formula is $[ML^2T^{-2}A^{-1}]$.

EXAMPLE 1. A long solenoid of radius 4 cm, length 400 cm carries a current of 3 A. The total number of turns is 100. Assuming ideal solenoid, find the flux passing through a circular surface having centre on axis of solenoid of radius 3 cm and is perpendicular to the axis of solenoid (i) inside and (ii) at the end of solenoid.

Sol.



Number of turns per unit length is given by

$$n = \frac{N}{l} = \frac{100}{4} = 25\ \text{turns/m}$$

(i) Magnetic field of a solenoid at a point inside is

$$B = \mu_0 ni$$

Area of cross-section of the solenoid,

$$A = \pi r_1^2 \text{ and } \theta = 0°$$

Magnetic flux,

$$\phi_B = BA\cos\theta$$
$$= \mu_0 ni\,\pi r_1^2\cos 0°$$
$$= 4\pi\times10^{-7}\times25\times3\times\pi\times(3\times10^{-2})^2$$
$$= 0.27\times10^{-6}\ \text{Wb}$$
$$= 0.27\ \mu\text{Wb}$$

(ii) At the end, magnetic field of solenoid is

$$B = \frac{1}{2}\mu_0 ni$$

$$\therefore\ \phi_B = \frac{0.27}{2} = 0.135\ \mu\text{Wb}$$

Experiments of Faraday and Henry

The discovery and understanding of electromagnetic induction are based on a long series of experiments carried out by Faraday and Henry.

It is the relative motion between the magnet and closed coil, which is responsible for generation or induction of electric current in the coil.

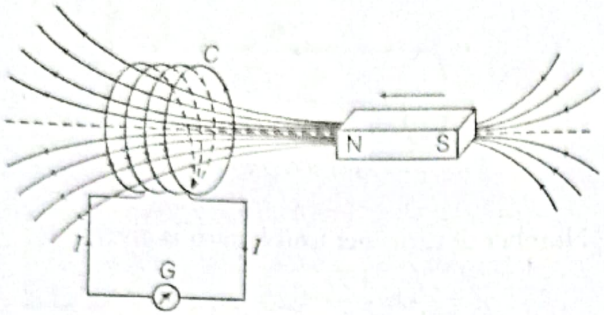
Whenever magnetic field linked with a closed coil changes, an emf is induced in the coil, which is called **induced emf.**

Some of the important experiments are given below

First Experiment (Current Induced by a Magnet)

Consider a coil C of few turns of conducting material insulated from one another and is connected to a sensitive galvanometer G.

Whenever there is a relative motion between the coil and magnet, the galvanometer shows a deflection indicating that current is induced in the coil.

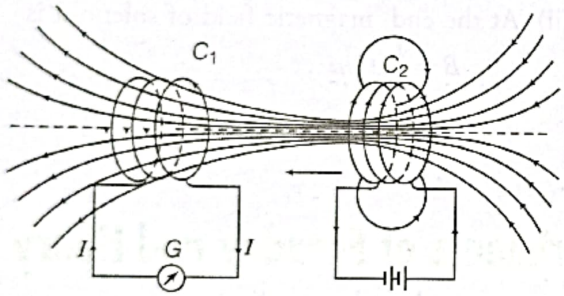


EMI with a stationary coil and moving magnet

Therefore, relative motion between the magnet and the coil generates electric current in the coil and the current generated is called **induced current.**

Second Experiment (Current Induced by a Current)

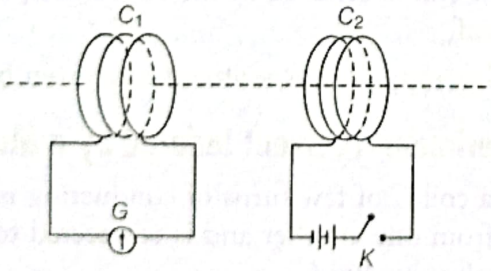
When the bar magnet is replaced by a second coil C_2 connected to a battery, the steady current in coil C_2 produces a steady magnetic field. As coil C_2 is moved towards coil C_1, the galvanometer shows a deflection. This indicates that electric current is induced in coil C_1.



EMI with one coil stationary and another moving

When coil C_2 is moved away, the galvanometer shows a deflection again but this time, in the opposite direction. The deflection lasts as long as coil C_2 is in the motion.

Third Experiment (Current Induced by Changing Current)



EMI with changing current in one coil

The figure shows two coils C_1 and C_2 held stationary. Coil C_1 is connected to galvanometer G, while the second coil C_2 is connected to a battery through a tapping key K.

It is observed that the galvanometer shows a momentary deflection when the tapping key K is pressed. If the key is pressed continuously, there is no deflection in the galvanometer. When the key is released, a momentary deflection is observed again but in opposite direction.

All experimental observations lead us to conclude that current is induced in the coil, whenever the amount of magnetic flux linked with the coil changes.

Note Presence of magnetic flux is not enough. The amount of magnetic flux linked with the coil must be change in order to produce an induced current in the coil.

Faraday's Laws of Electromagnetic Induction

On the basis of above experiments, following are the two laws of Faraday's electromagnetic induction stated as below

Faraday's First Law

Whenever the amount of magnetic flux linked with a circuit changes, an emf and consequently a current is induced in the circuit. It lasts so long as the change in magnetic flux continues.

Faraday's Second Law

The magnitude of the induced emf in a circuit is equal to the rate of change of magnetic flux linked with the circuit. Mathematically, Faraday's second law can be expressed as,

Induced emf $\propto$ Rate of change of magnetic flux

i.e. $$e = \frac{-d\phi_B}{dt}$$

The negative sign in above relation indicates that the induced emf always opposes the change in the magnetic flux through the circuit.

In the case of a closely wound coil of N turns, the change of flux associated with each turn is same. Therefore, the expression for the total induced emf is given by

$$e = -N\frac{d\phi_B}{dt}$$

The induced emf can be increased by increasing the number of turns N of a closed coil.

Induced Emf and Current

If N is the number of turns and R is the resistance of a coil, and the magnetic flux linked with its each turn changes by $d\phi$ in short time interval dt, then

Induced emf in the coil, $e = -N\frac{d\phi_B}{dt}$

Induced current flowing through the coil,

$$I = \frac{e}{R} = -\frac{N}{R}\cdot\frac{d\phi_B}{dt} \qquad \left[\because I = \frac{V}{R}\right]$$

Electric charge flows due to induced current,

$$q = Idt = -\frac{N}{R}d\phi_B$$

EXAMPLE 2. A magnetic field of flux density 10 T acts normally to the coil of 50 turns having 100 cm^2 area. Find emf induced, if the coil is removed from the magnetic field in 0.15 s.

Sol. Given, $B = 10$ T, $N = 50$ turns,

$A = 100\ cm^2 = 10^{-2} m^2$, $dt = 0.15$ s

Magnetic flux linked with the coil initially,

$\phi_1 = NBA = 50 \times 10 \times 10^{-2} = 5$ Wb

But magnetic flux linked with the coil finally, i.e. (when removed from the magnetic field), $\phi_2 = 0$.

$\therefore$ Emf induced, $e = \frac{-d\phi}{dt} = -\left(\frac{\phi_2 - \phi_1}{dt}\right)$

$= -\left(\frac{0-5}{0.15}\right)$

$= 33.33$ V

EXAMPLE 3. A square loop of side 10cm and resistance 0.5 Ω is placed vertically in the East-West plane. A uniform magnetic field of 0.10 T is set up across the plane in the North-East direction. The magnetic field is decreased to zero in 0.70 s at a steady rate. Determine the magnitudes of induced emf and current during this time interval.

Sol. Given, $B = 0.10$ T, $A = 100\ cm^2 = 10^{-2} m^2$, $\theta = 45°$ (as the angle made by area vector of the loop with magnetic field is 45°),

$R = 0.5\ \Omega$, $dt = 0.70$ s

Initial magnetic flux, $\phi_1 = BA\cos\theta$

$= 0.10 \times 10^{-2} \times \cos 45° \qquad \left(\because \cos 45° = \frac{1}{\sqrt{2}}\right)$

$= \frac{10^{-3}}{\sqrt{2}}$ Wb

As, final magnetic flux, $\phi_2 = 0$.

$\therefore$ Magnitude of induced emf is

$$e = -\frac{d\phi}{dt} = -\left(\frac{\phi_2 - \phi_1}{dt}\right) = -\left(\frac{0 - \frac{10^{-3}}{\sqrt{2}}}{0.70}\right)$$

$\approx 10^{-3}$ V $= 1.0$ mV

$\therefore$ Magnitude of induced current,

$$I = \frac{e}{R} = \frac{10^{-3}\text{ V}}{0.5\ \Omega} = 2 \times 10^{-3}\text{ A} = 2\text{ mA}$$

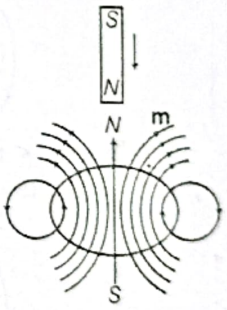
Lenz's Law

This law gives us the direction of induced current in the circuit.

According to this law, the polarity of emf induced is such that, it tends to produce a current which opposes the change in magnetic flux that produced it.

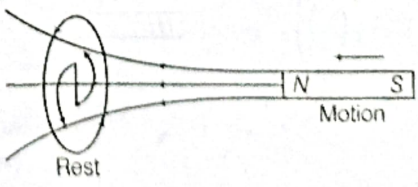
Illustration of Lenz's Law

As the magnet is moved towards the loop, a particular amount of current is induced in the loop. The magnetic field is produced by the current, with magnetic dipole moment m oriented, so as to oppose the motion of magnet.

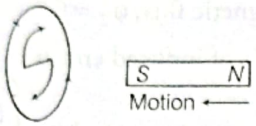


Thus, the induced current must be counter-clockwise as shown in the figure. When the North pole of a magnet moves towards a stationary loop, an induced current I flows in anti-clockwise sense as seen from the above, at which the magnet is located.

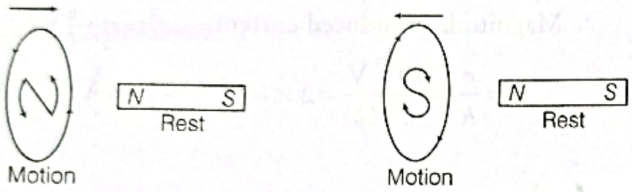
- The anti-clockwise sense corresponds to the generation of North pole which opposes the motion of the approaching N-pole of the magnet.



- When the North pole of the magnet is moved away from the loop, the current I flows in the clockwise sense, which corresponds to the generation of South pole as shown in figure. The induced South pole opposes the motion of the receding North pole.



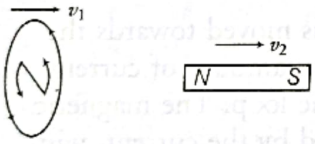
- The directions of current induced in all above cases remain same, if instead of the loop, the magnet is kept stationary and loop is moved towards or away from it.



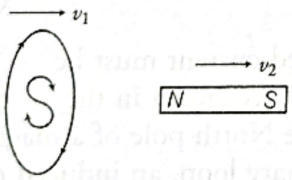
Relative Motion between Loop and Magnet

If both the loop and magnet are in relative motion w.r.t. each other, the induced pole on the loop facing magnet is according to Lenz's law

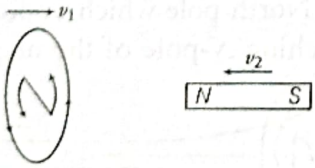
(i) When $v_1 > v_2$, i.e. the loop is approaching towards N-pole, hence, induced pole in loop is N-pole.



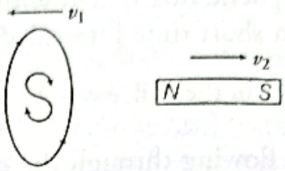
(ii) When $v_1 < v_2$, i.e. the loop is receding away from N-pole, hence induced pole in loop is S-pole.



(iii) When loop and magnet have opposite directions of velocities, and loop is approaching towards N-pole, then, induced pole in loop is N-pole.



(iv) When loop and magnet have opposite directions of velocities and loop is receding away from N-pole, then induced pole in loop is S-pole.

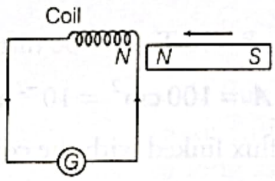


Thus, Lenz's law is used to find the direction of induced current in a closed circuit.

Lenz's Law and Conservation of Energy

Lenz's law is in accordance with the law of conservation of energy.

In the alongside circuit, when N-pole of magnet is moved towards the coil, the front face of the coil acquires North polarity. Thus, work has to be done against the force of repulsion in bringing the magnet closer to the coil.



When N-pole of magnet is moved away, South pole develops on the front face of the coil. Therefore, work has to be done against the force of attraction in taking the magnet away from the coil. This mechanical work in moving the magnet w.r.t. the coil changes into electrical energy producing induced current. Hence, energy transformation takes place. When we do not move the magnet, work done is zero. Therefore, induced current is also zero.

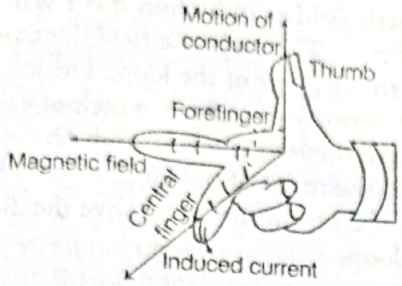
Hence, **Lenz's law obeys the law of conservation of energy.**

Fleming's Right Hand Rule

Fleming's right hand rule also gives the direction of induced current in a conductor moving in a magnetic field. According to this rule,

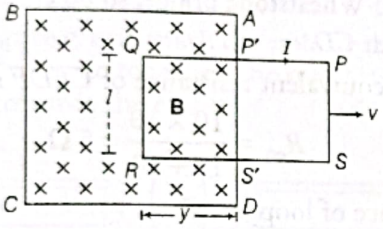
If we stretch the thumb, the forefinger and the central finger of right hand in such a way that all these three are mutually perpendicular to each other and if thumb represents the direction of motion of the conductor and the forefinger represents the direction of magnetic

field, then central finger will represent the direction of induced current as shown below.



Motional Electromotive Force

Consider a uniform magnetic field B confined to the region $ABCD$ and a coil $PQRS$ is placed inside the magnetic field. At any time t, the part $P'Q = S'R = y$ of the coil is inside the magnetic field. Let l be the length of the arm of the coil.



Inducing current by changing the area of the rectangular loop

Area of the coil inside the magnetic field at time t,

$$\Delta S = QR \times RS' = ly$$

Magnetic flux linked with the coil at any time t,

$$\phi = B\Delta S = Bly$$

If e is the induced emf, then according to Faraday's law,

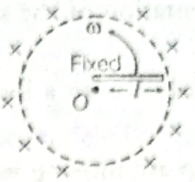
$$e = -\frac{d\phi}{dt} = -\frac{d(Bly)}{dt} = Bl\left(\frac{-dy}{dt}\right) = Blv$$

or $\boxed{e = Blv}$

where, $-\frac{dx}{dt} = v$ is the velocity of the coil.

From Fleming's right hand rule, the current due to induced emf will flow from the end R to Q, i.e. along $SRQP$ in the coil. This induced electromotive force (emf) Blv is called **motional emf.**

Note When a conducting rod of length l fixed at its one end moves on a circular path with angular velocity ω in a uniform magnetic field B normal to it, then induced emf produced in it is $e = \frac{1}{2}B\omega l^2$.



EXAMPLE 4. A wire of length 0.3 m moves with a speed of 20 m/s perpendicular to the magnetic field of induction 1 Wb/m^2 Calculate the induced emf.

Sol. Given, velocity, $v = 20$ m/s

Length, $l = 0.3$ m

Angle, $\theta = 90°$

Magnetic field, $B = 1$ Wb/m^2

As, induced emf, $e = Blv = 1 \times 0.3 \times 20 = 6$ V

EXAMPLE 5. A wheel with 10 metallic spokes each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of the earth's magnetic field H_E at a place. If $H_E = 0.4$ gauss at the place, what is the induced emf between the axle and the rim of the wheel? Take, 1 gauss $= 10^{-4}$ T.

Sol. Induced emf $= \frac{1}{2}B\omega l^2 = (1/2)\omega BR^2$

$$\left[\because \omega = 2\pi f = 2\pi\frac{120}{60} = 4\pi \text{ and } l = 2R\right]$$

$= (1/2) \times 4\pi \times 0.4 \times 10^{-4} \times (0.5)^2$

$= 6.28 \times 10^{-5}$ V

The number of spokes is immaterial because the emf's across the spokes are in parallel.

Methods of Producing Induced EMF

As we know that emf is induced in a circuit, whenever the amount of magnetic flux linked with circuit is changed.

$\because \phi = BA\cos\theta$, therefore, the magnetic flux ϕ can be changed by changing the value of B, A or θ. So, there are three methods of producing induced EMF, i.e.,

(i) By changing the magnitude of magnetic field B

(ii) By changing the area

(iii) By changing angle θ between the direction of B and normal to the surface area A, *i.e.*, changing the relative orientation of the surface area and magnetic field.

The magnetic flux linked with a loop does not change when

(i) magnet and loop are moving with the same velocity.

(ii) magnet is rotated around its axis without changing its distance from the loop.

(iii) loop is moved in a uniform magnetic field and the whole of the loop remains in the field.

Induced Quantities and Their Formulae

S.No.	Name of the quantity	Formula	Circuit open/ closed	Dependence upon resistance	SI unit
1.	Induced emf	$e = -\frac{d\phi}{dt}$	Open or closed	No	volt
2.	Induced current	$I = -\frac{d\phi}{Rdt}$	Closed	Yes	ampere
3.	Charge flown due to induced current	$dq = -\frac{d\phi}{R}$	Closed	Yes	coulomb

Induced Current in a Circuit

If R is the electrical resistance of the circuit, then induced current in the circuit is given by

$$I = \frac{e}{R}$$

If induced current is produced in a coil rotated in a uniform magnetic field, then

$$I = \frac{NBA\omega \sin \omega t}{R} = I_0 \sin \omega t$$

where, $I_0 = \frac{NBA\omega}{R}$ = peak value of induced current

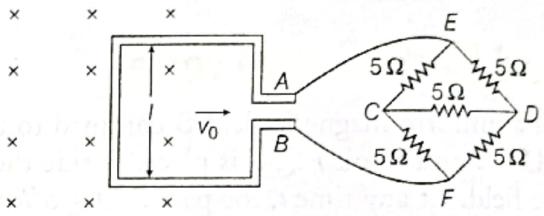
N = number of turns in the coil,

B = magnetic field,

ω = angular velocity of rotation

and A = area of cross-section of the coil.

EXAMPLE 6. A square metal wire loop of side 20 cm and resistance 2 Ω is moved with a constant velocity v_0 in a uniform magnetic field of induction $B = 1\ \text{Wb/m}^2$ as shown in the figure. The magnetic field lines are perpendicular to the plane of the loop. The loop is connected to a network of resistance each of value 5 Ω. The resistances of the lead wires BF and AE are negligible. What should be the speed of the loop, so as to have a steady current of 2 mA in the loop? Give the direction of current in the loop.



Sol. From the figure, we see that, network $CEDF$ is balanced Wheatstone bridge, so no current will flow in branch CD.

So, the equivalent resistance of $CEDF$ network is

$$R_{eq} = \frac{10 \times 10}{10 + 10} = 5\ \Omega$$

Resistance of loop = 2 Ω

$$R_{total} = 2 + R_{eq}$$
$$= 2 + 5 = 7\ \Omega$$

We know that, induced emf,

$$e = Bv_0 l$$

and induced current,

$$I = \frac{e}{R_{total}} = \frac{Bv_0 l}{7}$$

$$\Rightarrow 2 \times 10^{-3} = \frac{1 \times v_0 \times 0.2}{7}$$

$$\Rightarrow \quad v_0 = 7 \times 10^{-2}\ \text{m/s} = 7\ \text{cm/s}$$

As flux is decreasing, so induced current I will be clockwise.

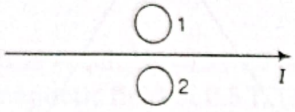
Topic Practice 1

Theoretical Questions

1. Two coils of wire A and B are placed mutually perpendicular. When a current induced is changed in any one coil, will the current induced in another coil?

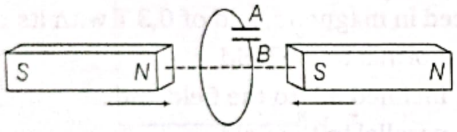
2. A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop? Justify. **(Delhi 2017)**

3. What is the direction of induced currents in metal rings 1 and 2, when current I in the wire is increasing steadily?



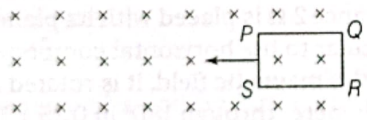
(All India 2017)

4. In the figure given, mark the polarity of plates A and B of a capacitor when the magnets are quickly moved towards the coil. **(All India 2017 C)**

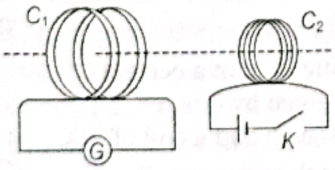


5. The closed loop $PQRS$ of wire is moved into a uniform magnetic field at right angles to the plane of the paper as shown in the figure. Predict the direction of the induced current in the loop.

(Foreign 2012)



6. A current is induced in coil C_1 due to the motion of current carrying coil C_2.



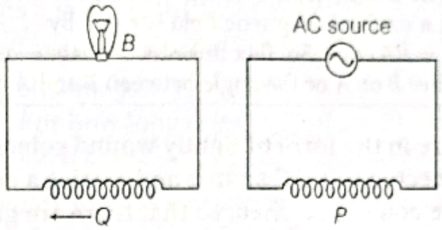
(i) Write any two ways by which a large deflection can be obtained in the galvanometer G.

(ii) Suggest an alternative device to demonstrate the induced current in place of a galvanometer. **(Delhi 2011)**

7. A coil Q is connected to low voltage bulb B and placed near another coil P as shown in the figure. Give reasons to explain the following observations.

(i) The bulb B lights.

(ii) Bulb gets dimmer, if the coil Q is moved towards left.



(Delhi 2010)

8. Two identical loops, one of copper and the other of aluminium are rotated with the same angular speed in the same magnetic field. Compare

(i) the induced emf and

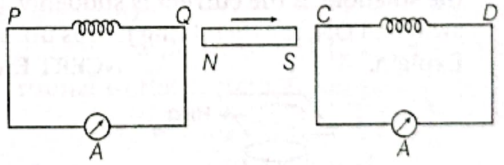
(ii) the current produced in the two coils. Justify your answer. **(All India 2010)**

9. State Lenz's law. A metallic rod held horizontally along East-West direction, is allowed to fall under gravity. Will there be an emf induced at its ends? Justify your answer.

(Delhi 2013)

10. A bar magnet is moved in the direction indicated by the arrow between two coils PQ and CD. Predict the directions of induced current in each coil.

(All India 2012)

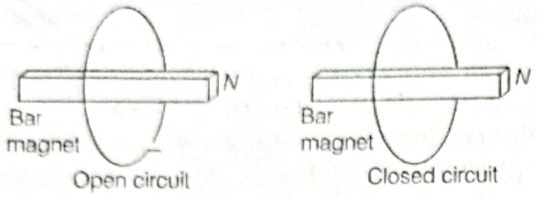


11. A rectangular loop of length l and breadth b is placed at distance of x from infinitely long wire carrying current i such that the direction of current is parallel to breadth. If the loop moves away from the current wire in a direction perpendicular to it with a velocity v, what will be the magnitude of emf in the loop?

12 Consider a magnet surrounded by a wire with an ON/OFF switch as shown in the figure. If the switch is thrown from the OFF position (open

circuit) to the ON position (closed circuit), will a current flow in the circuit? Explain.

(NCERT Exemplar)



Hints : The magnetic flux linked with a uniform surface area A in a uniform magnetic field is given by $\phi = \mathbf{B} \cdot \mathbf{A} = BA\cos\theta$. So, flux linked will change, only when either B or A or the angle between B and A changes.

13. A wire in the form of tightly wound solenoid is connected to a DC source and carries a current I. If the coil is stretched, so that there are gaps between successive elements of the spiral coil, will the current increase or decrease? Explain.

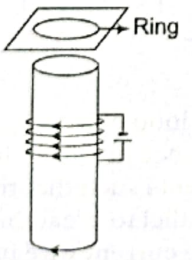
(NCERT Exemplar)

Hints: Here, the application of Lenz's law is tested through this problem.

14. A solenoid is connected to a battery, so that a steady current flows through it. If an iron core is inserted into the solenoid, will the current increase or decrease? Explain. **(NCERT Exemplar)**

15. (i) A metal ring is held horizontally and bar magnet is dropped through the ring with its length along the axis of the ring. What will be the acceleration of a falling magnet?

(ii) Consider a metal ring kept on top of a fixed solenoid (say on a cardboard) (see figure). The centre of the ring coincides with the axis of the solenoid. If the current is suddenly switched ON, the metal ring jumps up. Explain. **(NCERT Exemplar)**



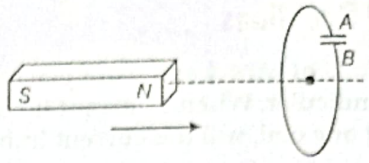
16. On what factors does the magnitude of induced emf in a coil depend?

17. If a coil is removed from a magnetic field

(i) slowly and

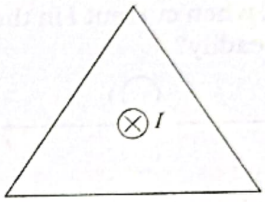
(ii) rapidly, then in which case, more work will be done?

18. In the given figure, a bar magnet is quickly moved towards a conducting loop having a capacitor. Predict the polarity of the plates A and B of the capacitor.



(All India 2014)

19. A current carrying straight wire passes inside a triangular coil as shown in figure. The current in the wire is perpendicular to paper inwards. Find the direction of the induced current in the loop, if current in the wire is increased.



Numerical Problems

20. A rectangular loop of area 20 cm × 30 cm is placed in magnetic field of 0.3 T with its plane

(i) normal to the field

(ii) inclined 30° to the field and

(iii) parallel to the field.

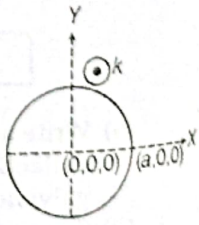
Find the flux linked with the coil in each case.

21. The magnetic flux through a coil perpendicular to the plane is given by $\phi = 5t^3 + 4t^2 + 2t$. Calculate induced emf through the coil at $t = 2$ s.

22. A circular coil of radius 10 cm, 500 turns and resistance 2 Ω is placed with its plane perpendicular to the horizontal component of the earth's magnetic field. It is rotated about its vertical diameter through 180° in 0.25 s. Estimate the magnitude of the emf and current induced in the coil. Horizontal component of the earth's magnetic field at the place is 3×10^{-5} T.

(NCERT Intext)

23. A magnetic field in a certain region is given by $\mathbf{B} = B_0\cos(\omega t)\hat{\mathbf{k}}$ and a coil of radius a with resistance R, is placed in the XY-plane with its centre at the origin in the magnetic field as shown in the figure. Find the magnitude and the direction of the current at $(a, 0, 0)$ at $t = \frac{\pi}{2\omega}, t = \frac{\pi}{\omega}$ and $t = \frac{3\pi}{2\omega}$.

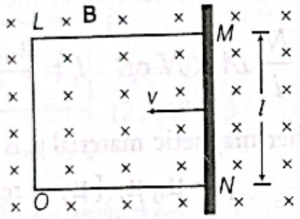


(NCERT Exemplar)

24. A horizontal straight wire 10 m long extending from East to West is falling with a speed of 5.0 m/s, at right angles to the horizontal component of the earth's magnetic field, 0.30×10^{-4} Wb/m^2.
(i) What is the instantaneous value of the emf induced in the wire?
(ii) What is the direction of the emf?
(iii) Which end of the wire is at the higher electrical potential? (NCERT)

25. A 1 m long conducting rod rotates with an angular frequency of 400 rad/s about an axis normal to the rod passing through its one end. The other end of the rod is in contact with a circular metallic ring. A constant magnetic field of 0.5 T parallel to the axis exists everywhere. Calculate the emf developed between the centre and the ring. (NCERT)

26. A rectangular conductor *LMNO* is placed in a uniform magnetic field of 0.5 T. The field is directed perpendicular to the plane of the conductor.



When the arm *MN* of length 20 cm is moved towards left with a velocity of 10 ms^{-1}, calculate the emf induced in the arm. Given, the resistance of the arm to be 5 Ω (assuming that other arms are of negligible resistance), find the value of the current in the arm. (All India 2013)

27. A rectangular loop of sides 8 cm and 2 cm with a small cut is moving out of a region of a uniform magnetic field of magnitude 0.3 T directed normal to the loop. What is the voltage developed across the cut, if velocity of loop is 1 cms^{-1} in a direction normal to the
(i) longer side?
(ii) shorter side of the loop?
For how long does the induced voltage last in each case? (NCERT)

28. A square loop of side 20 cm is initially kept 30 cm away from a region of a uniform magnetic field of 0.1 T as shown in the figure. It is then moved towards the right with a velocity of 10 cm s^{-1} till it goes out of the field. Plot a graph showing the variation of
(i) magnetic flux (ϕ) through the loop with time (t).
(ii) induced emf (ε) in the loop with time t.
(iii) induced current in the loop, if it has resistance of 0.1 Ω.

Self and Mutual Induction

Self-Inductance

It is the property of a coil by virtue of which, the coil opposes any change in the strength of current flowing through it by inducing an emf in itself. This induced emf is also called **back emf**. When the current in a coil is switched ON, it opposes the growth of the current and when the current is switched OFF, the self-induction opposes the decay of the current. So, self-induction is also called the **inertia of electricity**.

Coefficient of Self-Induction

Let us consider a coil of N turns carrying a current I. Let ϕ_B be the magnetic flux linked with each turn of the coil. Then, the number of flux linked through the coil will be $N\phi_B$.

If no magnetic materials (iron, etc.) are present near the coil, then the number of flux linkages with the coil is proportional to the current I, i.e.

$$N\phi_B \propto I \text{ or } N\phi_B = LI$$

where, L is constant called the **coefficient of self-induction** or **self-inductance** of the coil.
By the above equation, we have

$$\boxed{L = \frac{N\phi_B}{I}} \quad \ldots(i)$$

If $I = 1$, then $L = N\phi_B$.

Hence, the coefficient of self-induction of a coil is numerically equal to the number of magnetic flux linkages with the coil when unit current is flowing through the coil.

If on changing the current through the coil, the back emf induced in the coil be e, then by Faraday's law, we have

$$e = -N\frac{d\phi_B}{dt} = -\frac{d(N\phi_B)}{dt}$$

where, $\Delta(N\phi_B)/\Delta t$ is the rate of change of magnetic flux (due to change of current) in the coil. But $N\phi_B = LI$.

$$\therefore \quad e = -\frac{d(LI)}{dt} = -L\frac{dI}{dt}$$

where, dI/dt is the rate of change of current in the coil. The negative sign indicates that the induced emf e is always in such a direction that it opposes the change of current in the coil. From the above formula, we have

$$\boxed{L = -\frac{e}{dI/dt}} \quad ...(ii)$$

If $dI/dt = 1$, then $L = e$ (numerically).

Hence, the coefficient of self-induction of a coil is numerically equal to the emf induced in the coil when the rate of change of current in the coil is unity.

The SI unit of the coefficient of self-induction is henry (H) and its dimensions are $[ML^2T^{-2}A^{-2}]$.

Thus, the self-inductance of a coil is 1 henry when an induced emf of 1volt is set up in the coil due to a current changing at the rate of 1 ampere per second in the coil, i.e.

$$1 \text{ henry} = \frac{1 \text{ volt}}{1 \text{ ampere/second}}$$

Thus, as before, $1\ H = 1\ Vs\ A^{-1} = 1\ T\text{-}m^2/A$

$$= 1 Wb\ A^{-1}$$

But from Eq. (i), 1 henry = 1 weber/ampere.

The smaller units for L are millihenry (mH) and microhenry (μH).

$$1\ mH = 10^{-3}\ H,\ 1\ \mu H = 10^{-6}\ H$$

Self-Inductance of Long Solenoid

A long solenoid is one whose length is very large as compared to its area of cross-section. The magnetic field B at any point inside such a solenoid is practically constant and is given by

$$B = \frac{\mu_0 NI}{l} = \mu_0 nI \quad \left[\because n = \frac{N}{l}\right] ...(i)$$

where, μ_0 = magnetic permeability of free space,
N = total number of turns in the solenoid,
l = length of the solenoid
and n = number of turns per unit length.

∴ Magnetic flux through each turn of the solenoid,

$\phi = B \times$ area of the each turn

$$\phi = \left(\mu_0 \frac{N}{l} I\right) A$$

where, A = area of each turn of the solenoid.

Total magnetic flux linked with the solenoid

= Flux through each turn × Total number of turns

$$N\phi = \mu_0 \frac{N}{l} IA \times N \quad ...(ii)$$

If L is coefficient of self-inductance of the solenoid, then

$$N\phi = LI \quad ...(iii)$$

From Eqs. (ii) and (iii), we get

$$LI = \mu_0 \frac{N}{l} IA \times N \text{ or } \boxed{L = \frac{\mu_0 N^2 A}{l}}$$

If core of any other magnetic material μ is placed, then

$\mu = \mu_0 \mu_r$ [μ_r = relative magnetic permeability]

$$\therefore \quad \boxed{L = \frac{\mu_0 \mu_r N^2 A}{l}}$$

EXAMPLE 1. Current in a circuit falls steadily from 2.0 A to 0.0 A in 10 ms. If an average emf of 200 V is induced, calculate the self-inductance of the circuit. **(Foreign 2011)**

Sol. Given, $\Delta I = -2$ A, $\Delta t = 10 \times 10^{-3}$ s

$e = 200$ V, $L = ?$

∵ Induced emf, $e = -L\frac{\Delta I}{\Delta t}$

$$\Rightarrow \quad 200 = -L\left(\frac{-2}{10 \times 10^{-3}}\right)$$

$$\Rightarrow \quad 200 = L \times 2 \times 10^2$$

∴ Self-induction, $L = 1$ H

EXAMPLE 2. What is the self-inductance of a solenoid of length 40 cm, area of cross-section 20 cm^2 and total number of turns is 800?

Sol. Given, $l = 40\text{ cm} = 0.4\text{ m}$

$A = 20\text{ cm}^2 = 20 \times 10^{-4}\text{ m}^2$

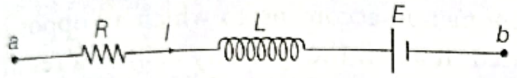
$N = 800,\ L = ?$

$$\therefore \quad L = \frac{\mu_0 N^2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times (800)^2 \times 20 \times 10^{-4}}{0.4}$$

$$= 4.02 \times 10^{-3}\text{ H}$$

EXAMPLE 3. In the circuit diagram shown in figure, $R = 10\ \Omega$, $L = 5$ H, $E = 20$ V, $I = 2$ A. This current is decreasing at a rate of -1.0 A/s. Find V_{ab} at this instant.



Sol. Potential difference across inductor is given as

$$V_L = L\frac{dI}{dt} = (5)(-1.0) = -5\text{ V}$$

Now, using Kirchhoff's second law,

$V_a - IR - V_L - E = V_b$

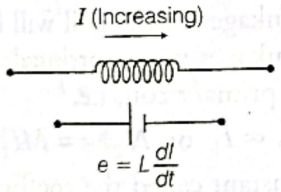
$$\therefore \quad V_{ab} = V_a - V_b = E + IR + V_L$$

$$= 20 + (2)(10) - 5$$

$$= 35\text{ V}$$

Energy Stored in an Inductor

The energy of a capacitor is stored in the electric field between its plates. Similarly, an inductor has the capability of storing energy in its magnetic field.



An increasing current in an inductor causes an emf between its terminals which is given by

$$e = L \cdot \frac{dI}{dt}$$

Work required to move a charge dq against emf e is

$$dW = edq = L \cdot \frac{dI}{dt} \cdot dq = L\left(\frac{dq}{dt}\right) \cdot dI = LIdI$$

$\therefore$ Total work to build up current from zero to maximum value I.

$$W = \int_0^W dW = \int_0^I LI\,dI = \frac{1}{2}LI^2$$

This work is stored as potential energy U in the inductor.

$$\therefore \quad \boxed{W = U = \frac{1}{2}LI^2}$$

Energy stored per unit volume(V) in magnetic field is known as **energy density.**

$$\therefore \quad \boxed{\text{Energy density} = \frac{U}{V} = \frac{1}{2}\frac{B^2}{\mu_0}}$$

Thus, if $I = 1$ A, then $2W = L$

Hence, the coefficient of self-inductance is equal to twice the work done in establishing a flow of one ampere current in the circuit.

EXAMPLE 4. Two coils having self-inductances, $L_1 = 5$ mH and $L_2 = 1$ mH. The current in the coil is increasing at same constant rate at a certain instant and the power supplied to the coils is also same. Find the ratio of

(i) induced voltages

(ii) currents

(iii) energy stored in two coils at that instant.

Sol. Given, $L_1 = 5$ mH and $L_2 = 1$ mH

(i) As we know, induced voltage is given by

$$e = \frac{Ldi}{dt}$$

$$\Rightarrow \quad \frac{e_1}{e_2} = \frac{L_1(dI/dt)}{L_2(dI/dt)} = \frac{L_1}{L_2} = \frac{5}{1} = 5:1 \qquad \ldots(i)$$

(ii) Power in the coil is given by

$$P = eI$$

Here, $P_1 = P_2 \quad \Rightarrow \quad e_1 I_1 = e_2 I_2$

$$\Rightarrow \quad \frac{I_1}{I_2} = \frac{e_2}{e_1}$$

Using Eq. (i), we can write as, $\frac{i_1}{i_2} = \frac{1}{5} = 1:5$

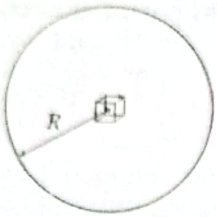
(iii) Energy stored in a coil is given by $U = \frac{1}{2}Li^2$

$$\therefore \quad \frac{U_1}{U_2} = \frac{(1/2)L_1 I_1^2}{(1/2)L_2 I_2^2} = \frac{L_1}{L_2}\left(\frac{e_2}{e_1}\right)^2$$

$$= \frac{5}{1}\left(\frac{1}{5}\right)^2 = 1:5$$

EXAMPLE 5. Suppose a cube of volume 2 mm^3 is placed at the centre of a circular loop of radius 5 cm carrying current 2 A. Find the magnetic energy stored inside the cube.

Sol.



Magnetic field at the centre of the circular loop is given by $B = \frac{\mu_0 i}{2R}$.

We know, energy density, $\mu = \frac{B^2}{2\mu_0}$ and energy stored in the cube will be given by

$$U = \mu V_0 = \frac{B^2}{2\mu_0} V_0$$

where, V_0 is the volume of the cube.

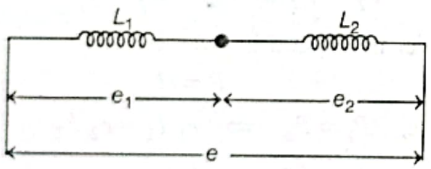
Substituting the values in the above equation,

$$= \frac{1}{2\mu_0} \times \left(\frac{\mu_0 I}{2R}\right)^2 V_0 = \frac{\mu_0 I^2 V_0}{8R^2}$$

$$= \frac{4\pi \times 10^{-7} \times 2^2 \times 2 \times 10^{-9}}{8 \times (0.05)^2} = 16\pi \times 10^{-14} \text{ J}$$

Grouping of Coils

(a) **Coils in Series** Suppose the two coils of inductance L_1 and L_2 are connected in series and the coefficient of coupling $K = 0$.

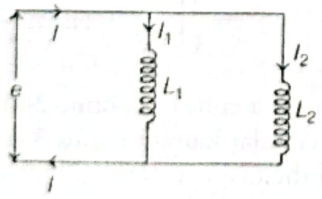


Since, current through each coil is same and potential divides, therefore, $e = e_1 + e_2$

$$\therefore \quad L_s \frac{dI}{dt} = L_1 \frac{dI}{dt} + L_2 \frac{dI}{dt}$$

$$\Rightarrow \quad \boxed{L_s = L_1 + L_2}$$

(b) **Coils in Parallel** Suppose two coils of inductance L_1 and L_2 are connected in parallel and the coefficient of coupling, $K = 0$.



Since, potential remains the same in parallel and current divides.

$$I = I_1 + I_2$$

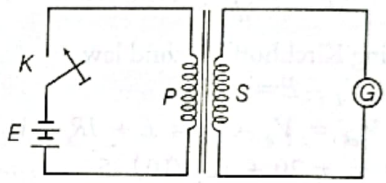
$$\therefore \quad \frac{dI}{dt} = \frac{dI_1}{dt} + \frac{dI_2}{dt}$$

Using $e = \frac{LdI}{dt}, \frac{e}{L_p} = \frac{e_1}{L_1} + \frac{e_2}{L_2}$

$$\Rightarrow \quad \boxed{\frac{1}{L_p} = \frac{1}{L_1} + \frac{1}{L_2}} \qquad \{\because e_1 = e_2 = e\}$$

Mutual Inductance

The phenomenon according to which an opposing emf is produced in a coil (i.e. primary coil) as a result of change in current or magnetic flux linked with a neighbouring coil (i.e. secondary coil) is called **mutual induction.**



Mutual induction

Coefficient of Mutual Induction

Let a current of I_1 ampere flows in the primary coil (P). Due to this current, the magnetic flux linked with each turn of the secondary coil (S) be ϕ_2. If N_2 is the number of turns in the secondary coil, then the number of flux linkages in the coil will be $N_2\phi_2$. This number of flux linkages is proportional to the current I_1 flowing in the primary coil, i.e.

$$N_2\phi_2 \propto I_1 \text{ or } N_2\phi_2 = MI_1$$

where, M is a constant called the **coefficient of mutual induction** or **mutual inductance** of the two coils. From the above equation, we have

$$M = \frac{N_2\phi_2}{I_1}$$

If $I_1 = 1$, then $M = N_2\phi_2$

Hence, the coefficient of mutual induction of two coils is equal to the number of magnetic flux linkages in one coil when a unit current flows in the other.

If on changing the current in the primary coil, the emf induced in the secondary coil is e_2, then according to Faraday's law, we have

$$e_2 = -N_2 \frac{d\phi_2}{dt} = -\frac{d(N_2\,\phi_2)}{dt}$$

where, $d\phi_2 / dt$ is the rate of change of magnetic flux in the secondary coil (due to change of current in the primary coil). But $N_2\phi_2 = MI_1$.

$$\therefore \quad e_2 = -\frac{d(MI_1)}{dt} = -M\frac{dI_1}{dt}$$

where, dI_1 / dt is the rate of change of current in the primary coil. The negative sign indicates that the direction of emf induced in the secondary coil is always such that it opposes any change in current in the primary coil.

From the above expression, we have

$$\boxed{\text{Mutual inductance, } M = -\frac{e_2}{dI_1 / dt}}$$

If $dI_1 / dt = 1$, then $M = e_2$ (numerically).

Hence, the coefficient of mutual induction of two coils is equal to the numerical value of the induced emf in one coil when the rate of change of current in other coil is unity.

The SI unit of the coefficient of mutual inductance is henry (H). Thus, **the mutual inductance of two coils is 1 henry when an induced emf of 1 volt is set up in one of them due to a current changing at the rate of 1 ampere per second in the other.**

i.e. $$1 \text{ henry} = \frac{1 \text{volt}}{1 \text{ ampere/second}}$$

or $$1\text{H} = 1\text{V s A}^{-1}$$

$$= 1 \text{ Wb A}^{-1}$$

The coefficient of mutual induction or mutual inductance of two coils depend on

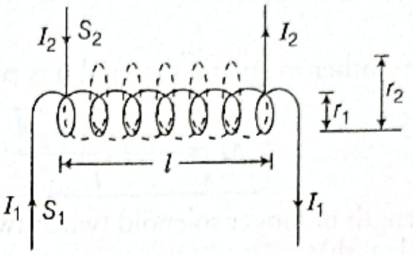
(i) geometry of two coils, i.e. size of coils, their shape, number of turns, nature of material on which two coils are wound.

(ii) distance between two coils.

(iii) relative placement of two coils (i.e. orientation of the two coils).

Mutual Inductance of Two Long Coaxial Solenoids

Consider two long solenoids S_1 and S_2 of same length l, such that solenoid S_2 surrounds solenoid S_1 completely.



Two long coaxial solenoids of same length l

Let n_1 be the number of turns per unit length of S_1, n_2 be the number of turns per unit length of S_2, I_1 be current passed through solenoid S_1 and ϕ_{21} be the flux linked with S_2 due to current flowing through S_1.

$$\phi_{21} \propto I_1 \text{ or } \phi_{21} = M_{21}\, I_1$$

where, M_{21} is the coefficient of mutual induction of the solenoid S_2 w.r.t. solenoid S_1.

When current is passed through solenoid S_1, an emf is induced in solenoid S_2.Magnetic field produced inside solenoid S_1 on passing current through it.

$$B_1 = \mu_0 n_1 I_1$$

Magnetic flux linked with each turn of solenoid S_2 will be equal to B_1 times the area of cross-section of solenoid S_1.

Magnetic flux linked with each turn of the solenoid $S_2 = B_1 A$.

Therefore, total magnetic flux linked with the solenoid S_2 will be

$$\phi_{21} = B_1 A \times n_2 l$$
$$= \mu_0 n_1 I_1 \times A \times n_2 l$$
$$\phi_{21} = \mu_0 n_1 n_2 A I_1 l$$
$$M_{21} = \mu_0 n_1 n_2 Al \quad \ldots(i)$$

Similarly, the mutual inductance between the two solenoids, when current is passed through solenoid S_2 and induced emf is produced in solenoid S_1 and is given by

$$M_{12} = \mu_0 n_1 n_2 Al$$
$$M_{12} = M_{21} = M \quad \text{(say)}$$

Hence, coefficient of mutual induction between two long solenoids, $M = \mu_0 n_1 n_2 Al$

We can rewrite Eq. (i) as,

$$M = \mu_0 \left(\frac{N_1}{l}\right)\left(\frac{N_2}{l}\right)\pi r_1^2 \times l$$

$$= \frac{\mu_0 N_1 N_2 A}{l}$$

If core of any other magnetic material μ is placed, then

$$\boxed{M = \frac{\mu_0 \mu_r N_1 N_2 A}{l}}$$

where, l = length of longer solenoid (when two solenoids of different length)

A = Area of cross-section of inner solenoid

EXAMPLE 6. There are two coils, which have mutual inductance of 10 H. When the circuit is closed, current in the primary coil is raised to 3 A within a time range of 1 millisecond. Calculate the emf induced in secondary coil.

Sol. Given, mutual inductance, $M = 10$ H

Change in current, $dI = 3$ A

Change in time, $dt = 1$ millisecond $= 10^{-3}$ s

emf induced, $e = ?$

emf induced in secondary coil is given by $e = \frac{M\,dI}{dt}$

$\therefore$ emf induced $= \frac{10 \times 3}{10^{-3}} = 3 \times 10^4$ V

EXAMPLE 7. A 1 m long solenoid with diameter 2 cm and 2000 turns has a secondary coil of 1000 turns wound closely near its mid-point. What will be the mutual inductance between the two coils?

Sol. Given, $l = 1$ m, $r = \frac{2}{2}$ cm $= 1$ cm $= 10^{-2}$ m

$N_1 = 2000, N_2 = 1000, A = \pi r^2 = \pi (10^{-2})^2$ m^2

$= \pi \times 10^{-4}$ m^2, $M = ?$

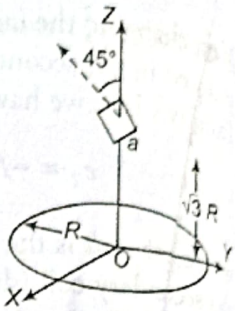
$\therefore$ Mutual inductance between the two coils is given by

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times 2000 \times 1000 \times \pi \times 10^{-4}}{10^{-2}}$$

$$= 78.9 \times 10^{-3} \text{ H}$$

EXAMPLE 8. A circular wire loop of radius R is placed in the XY-plane centred at the origin O. A square loop of side a ($a << R$) having two turns is placed with its centre at $z = \sqrt{3}\,R$ along the axis of the circular wire loop, as shown in figure. The plane of the square loop makes an angle of 45° with respect to the Z-axis. If the mutual inductance between the loops is given by $\frac{\mu_0 a^2}{2^{p/2} R}$, then find the value of p.



Sol. If I current flows through the circular loop, then magnetic field at the location of square loop is

$$B = \frac{\mu_0 I R^2}{2(R^2 + Z^2)^{3/2}}$$

Substituting the value of $Z = \sqrt{3}R$, we have

$$B = \frac{\mu_0 I}{16R}$$

Now, total flux through the square loop is

$$\phi_T = NBS \cos\theta$$

$$= (2)\left(\frac{\mu_0 I}{16R}\right) a^2 \cos 45°$$

Mutual inductance,

$$M = \frac{\phi_T}{I} = \frac{\mu_0 a^2}{2^{7/2} R} \quad \ldots(i)$$

On comparing the Eq. (i) with given,

$$M = \frac{\mu_0 a^2}{2^{p/2} R}$$

we get

$\therefore$ $p = 7$

AC Generator

It is a device which is used to convert mechanical energy into electrical energy.

Principle

It is based on the phenomenon of electromagnetic induction. The emf is induced in the rotating coil as flux is changing due to angle θ between B and A variation.

Main Parts of an AC Generator

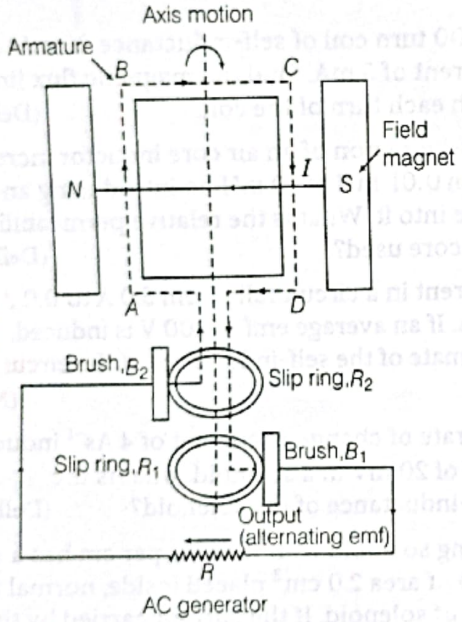
Main parts of an AC generator are shown in the figure and discussed as given below.

Armature A rectangular coil $ABCD$ consisting of a large number of turns of copper wire wound over a soft iron core is called **armature**. The soft iron core is used to increase the magnetic flux.

Field magnets Two pole pieces of a strong electromagnet.

Slip rings The ends of the coil $ABCD$ are connected to two hollow metallic rings R_1 and R_2.

Brushes B_1 and B_2 are two flexible metal plates or carbon rods. They are fixed and are kept in slight contact with R_1 and R_2.



AC generator

Theory and Working

As the armature of coil is rotated in uniform magnetic field, the magnetic flux linked with the coil changes and an emf is induced in the coil. Hence current flow in it.

Suppose the coil is rotating clockwise and is horizontal of some instance of time. At this instant, the arm AB of the coil is moving up and the arm CD is moving down.

According to Fleming's right hand rule, the current flows from A to B in the arm AB and C to D in the arm CD.

The brush B_1 at which the current is entering the external circuit is positive relative to the brush B_2 at which the current is leaving the external circuit.

During the time the coil rotates to the vertical position, the current in the armature will flow in the direction $ABCD$.

Now the coil $ABCD$ reaches the vertical position, the arms AD and CD move momentarily parallel to the field. So, in vertical position, the induced emf (current) is zero.

After half the rotation of coil, the arm AB moves down and the arm CD moves up. The current flow from B to A in arm AB and D to C in the arm CD.

The brush B_2 is now positive relative to B_1 and remains so long as the coil reaches the vertical position again.

After this B_1 is again positive relative to B_2.

Hence, induced current in the external circuit changes direction after every half rotation of the coil. Hence the current induced is alternating in a nature.

In commercial generators, the mechanical energy required for rotation of armature is provided by water falling from height, e.g. dams. These are called **hydroelectric generators.**

If the steam at high pressure is used to produce the rotation of armature, these are called **thermal generators.** Instead of coal, if a nuclear fuel is used, we get **nuclear power generators.**

Note In India, the frequency of generation of AC is 50 Hz.

EXAMPLE 9. An AC generator consists of a coil of 1000 turns and cross-sectional area of 100 cm^2, rotating at an angular speed of 100 rpm in a uniform magnetic field of 3.6×10^{-2} T. Calculate the maximum emf produced in the coil.

Sol. Given, $N = 1000$, $A = 100\,cm^2 = 10^{-2}\,m^2$,

$$\nu = 100 \text{ rpm} = \frac{100}{60} \text{ rps},$$

$$B = 3.6 \times 10^{-2}\,T,\ e_0 = ?$$

$\therefore$ Maximum emf produced in the coil is

$$e_0 = NBA\omega = NBA(2\pi\nu)$$

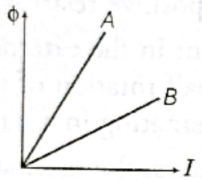
$$= 1000 \times 3.6 \times 10^{-2} \times 10^{-2} \times 2 \times \frac{22}{7} \times \frac{100}{60}$$

$$= 3.77\ V$$

Topic Practice 2

Theoretical Questions

1. Can a straight wire act as an inductor?
2. Define the term self-inductance of a coil. Write its SI unit. **(Delhi 2015)**
3. How can the self-inductance of a given coil having N number of turns, area of cross-section A and length l be increased? **(Foreign 2009; Delhi 2012)**
4. A plot of magnetic flux (ϕ) *versus* current (I) is shown in the figure for two inductors A and B. Which of the two has larger value of self-inductance? **(Delhi 2010)**



5. Self-induction is called the inertia of electricity. Why?
6. Define mutual inductance. Write its SI unit. **(Delhi 2016)**
7. How does the mutual inductance of a pair of coils change when
   (i) distance between the coils is increased and
   (ii) number of turns in the coils is increased? **(All India 2013)**
8. A source of emf e is used to establish a current I through a coil of self-inductance L. Show that the work done by the source to build up the current I is $\frac{1}{2}LI^2$. **(Delhi 2010 C)**
9. Define mutual inductance between two long coaxial solenoids. Find out the expression for the mutual inductance of inner solenoid of length l having the radius r_1 and the number of turns n_1 per unit length due to the second outer solenoid of same length and n_2 number of turns per unit length. **(Delhi 2012)**
10. Two concentric circular coils, one of radius r and the other of radius R are placed coaxially with their centres coinciding. For $R >>> r$, obtain an expression for the mutual inductance of the arrangement. **(Delhi 2011)**
11. A small square loop of wire of side l is placed inside a large square loop of wire of side $L (L >> l)$. The loops are coplanar and their centres coincide. Give the dependence of mutual inductance.
12. How can self-inductance of a given coil having N number of turns be changed, if N is doubled keeping other factors constant?
13. If two coils are very tightly wound over one another, will their mutual inductance increase or decrease as compared to the case when the coils are placed some distance apart?

Numerical Problems

14. A 200 turn coil of self-inductance 30 mH carries a current of 5 mA. Find the magnetic flux linked with each turn of the coil. **(Delhi 2011)**
15. Self-induction of an air core inductor increases from 0.01 mH to 10 mH on introducing an iron core into it. What is the relative permeability of the core used? **(Delhi 2011)**
16. Current in a circuit falls from 5.0 A to 0.0 A in 0.1 s. If an average emf of 200 V is induced, give an estimate of the self-inductance of the circuit. **(NCERT)**
17. If a rate of change of current of 4 As^{-1} induces an emf of 20 mV in a solenoid, what is the self-inductance of the solenoid? **(Delhi 2010)**
18. A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside, normal to the axis of solenoid. If the current carried by the solenoid changes steadily from 2 A to 4 A in 0.1 s, what is the induced voltage in the loop, while the current is changing? **(NCERT)**
19. A coil has a self-inductance of 10 mH. What is the maximum magnitude of the induced emf in the inductor, when a current $I = 0.1 \sin 200\,t$ ampere is sent through it?
20. A solenoid of radius 3 cm and length 1 m has 600 turns per metre. Calculate its self-inductance.
21. The current flowing in the two coils of self-inductance $L_1 = 16$ mH and $L_2 = 12$ mH are increasing at the same rate. If the power supplied to the two coils are equal, find the ratio of

(i) induced voltages (ii) the currents and
(iii) the energies stored in the coil at a given instant.
(Foreign 2014)

22. A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 to 20 A in 0.5 s, what is the change of flux linkage with the other coil? **(Delhi 2016)**

23. Two coils have mutual inductance of 1.5 H. If current in primary coil is raised to 5 A in one millisecond after closing the circuit, what is the emf induced in the secondary coil?

24. There are two coils A and B separated by some distance. If a current of 2 A flows through A, a magnetic flux of 10^{-2} Wb passes through B (no current through B). If no current passes through A and a current of 1 A passes through B, what is the flux through A? **(NCERT Exemplar)**

25. The flux linked with a large circular coil of radius R is 0.5×10^{-3} Wb. When a current of 0.5 A flows through a small neighbouring coil of radius r, calculate the coefficient of mutual inductance for the given pair of coils. If the current through the small coil suddenly falls to zero, what would be its effect in the larger coil? **(Delhi 2008)**

26. An AC generator consists of coil of 100 turns and cross-sectional area of 3 m^2, rotating at a constant angular speed of 60 rad s^{-1} in a uniform magnetic field 0.04T. The resistance of the coil is 500 Ω. Calculate (i) maximum current drawn from the generator and (ii) maximum power dissipation of the coil.